

Week 8 Report: Model Evaluation and Results Analysis

Project: Determining the Role of Big Data Applications in Fraud Detection and Financial Security in Financial Institutions of Ireland

Student: Yashwanth

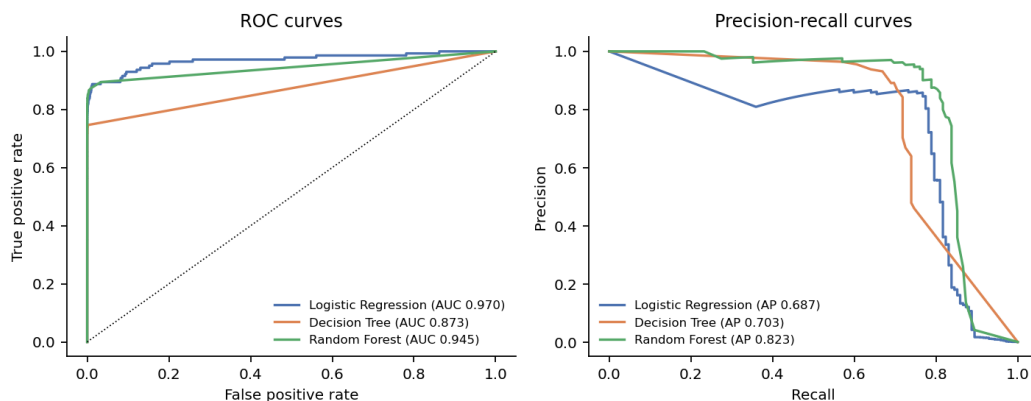
Supervisor: Dr. Prasanth Nayak

What I did this week

This week I evaluated the three models with class weights, as planned in my proposal, on the 85,118 test transactions. Random Forest is the best model: F1 0.818 (95% interval 0.766 to 0.869), PR-AUC 0.823, and only 4 false alarms. Logistic Regression finds the most frauds but raises 2,264 false alarms.

McNemar's test shows that Random Forest is significantly different from both other models (p below 0.001). Logistic Regression has the highest ROC-AUC but a low PR-AUC, which shows why PR-AUC is the better measure for rare fraud.

Model	Precision	Recall	F1	PR-AUC	False alarms
Random Forest	0.962	0.711	0.818	0.823	4
Decision Tree	0.461	0.746	0.570	0.703	124
Logistic Regression	0.053	0.887	0.100	0.687	2,264



ROC and precision-recall curves on the test set.

Next week

Analyse thresholds, robustness over time and which features drive the predictions.